



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



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Computational Fluid Dynamics

INTRODUCTION TO NUMERICAL METHODS FOR ASTROPHYSICS

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Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Fluid Elements

- *Fluids*: things that flow, conventionally liquids and gases, contrast solids with rigid lattice
- *Elements*: region with definable local variables, fulfilling certain conditions
 - small enough to ignore systematic variations, size below any characteristic scale length

$$\ell \ll \left| \frac{q}{\nabla q} \right|$$

- large enough to ignore discreteness noise, statistically sufficient number of particles

$$n\ell^3 \gg 1$$

- in collisional fluids, large enough to communicate local conditions, requires interactions

$$\ell \gg \lambda$$

Describing Fluid Equations

- *Eulerian View*: cells of small volumes in fixed grid, flow is defined for all $\mathbf{t}$ at position $\mathbf{r}$ with rate of change $\partial/\partial\mathbf{t}$ for any measurable quantity Q
- *Lagrangian View*: follow particular fluid elements, comoving reference system with flow evaluated in each one via $D/D\mathbf{t}$ as the rate of change
- *Translating Relation*: add convective derivative term to account for movement of fluid element

$$\frac{DQ}{Dt} = \frac{\partial Q}{\partial t} + \mathbf{u} \cdot \nabla Q$$

Continuous Variables

■ *Primitive Fields:*

- velocity $\mathbf{u}$ (vector)
- density ρ (scalar)
- pressure p (scalar)
- temperature T (scalar)

■ *Conserved Quantities:*

- mass M (scalar)
- energy E (scalar)
- momentum $\mathbf{P}$ (vector)

■ *Derived Values:*

- viscosity η (scalar)
- vorticity $\boldsymbol{\omega}$ (vector)

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Dimensionless Numbers

- *Reynolds Number*: ratio of inertial to viscous forces

$$R = \frac{\rho u L}{\eta} = \frac{u L}{\nu}$$

- $R < 1000$: Laminar Flow (reversible and ordered stream)
 - ▶ $R < 1$: Stokes Regime (without any separation)
 - ▶ $10 \lesssim R \lesssim 40$: Separation Zone (stable and stationary vortices)
- $R > 1000$: Turbulent Flow (stochastic and chaotic stream)
 - ▶ $40 \lesssim R \lesssim 4000$: Transitional Kármán Street (vortex shedding)
 - ▶ $R > 4000$: Kolmogorov Energy Cascade (fully developed phase)

Dimensionless Numbers

- *Mach Number*: ratio of bulk velocity magnitude to local sound speed

$$M = \frac{u}{c}$$

- $M \lesssim 0.3$: Incompressible (negligible density variations, pressure signals faster than flow)
- $M \approx 1$: Transonic (local excess of sound speed, weak shocks create drag)
- $M > 1$: Supersonic (thin front of discontinuity, bow shock heating)
- $M > 3$: Hypersonic (ram pressure stripping, dissociation and ionization)

Bulk Transport

- *Continuity Equation*: density ρ must increase where a fluid is compressed

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

- *Linear Advection*: velocity field $\mathbf{v}$ is free of divergence for incompressible fluids

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{u} = 0$$

- *Nonlinear Advection*: velocity field $\mathbf{u}$ is carried by itself instead of fixed background flow

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = 0$$

Linear Advection

Nonlinear Advection

Stochastic Transport

- *Diffusion*: kinematic viscosity ν describes spread due to random motion

$$\frac{\partial u}{\partial t} = \nu \nabla^2 u$$

- *Bateman-Burgers Equation*: combines shock generation with counteracting diffusive term

$$\frac{\partial u}{\partial t} + (u \cdot \nabla)u = \nu \nabla^2 u$$

Diffusion

Bateman-Burgers Equation

Equilibrium Search

- *Laplace Equation*: maximally smooth solution dictated solely by boundary conditions

$$\nabla^2 p = 0$$

- *Poisson Equation*: extension for steady state to accomodate internal sources and sinks

$$\nabla^2 p = f(r)$$

- Gravity: $\nabla^2 \Phi = 4\pi G\rho$
- Electrostatics: $\nabla^2 V = -\rho/\epsilon$
- Pressure: $\nabla^2 p = \rho \nabla \cdot ((u \cdot \nabla)u)$

Laplace Equation

Poisson Equation

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Lid Driven Cavity Flow

Pressure Driven Channel Flow

Nonuniform Periodic Flow

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Setup

Visualization

Overview

Physical Background

Fundamental Definitions

Basic Equations

Examples

Numerical Approaches

Finite Difference Method

Finite Volume Method

Finite Element Method

Practical Implementation

Kelvin-Helmholtz Instability

Rayleigh-Taylor Instability

Outlook

Setup

Visualization

Overview

Physical Background

- Fundamental Definitions
- Basic Equations
- Examples

Numerical Approaches

- Finite Difference Method
- Finite Volume Method
- Finite Element Method

Practical Implementation

- Kelvin-Helmholtz Instability
- Rayleigh-Taylor Instability
- Outlook

Outlook

■ Frameworks:

- Smoothed Particle Hydrodynamics (SPH)
- Lattice Boltzmann Method (LBM)
- Adaptive Mesh Refinement (AMR)

■ Regimes:

- *Plasma*: Magnetohydrodynamics (MHD)
- *Relativity*: Relativistic Hydrodynamics (RHD)
- *Degeneracy*: Quantum Effects
- *Collisionless*: Initial Distributions

■ Applications:

- Stellar Evolution
- Accretion Disks
- Cosmic Filaments

Thank You!

- C. Clarke & B. Carswell, *Principles of Astrophysical Fluid Dynamics* (Cambridge University Press) 2007
- L. A. Barba, *CFD Python: 12 Steps to Navier-Stokes*¹ 2022
- P. Mocz, *Medium*² 2020-2025

Resources on GitHub: [here](#)